

Practice Quiz: Systems -- The Interconnected Web of Your Life

Part A: Multiple Choice

1. A **system** in Active Inference is best defined as: A) Any group of people in the same location B) A collection of interacting components separated from the environment by a boundary C) A set of rules written down by an authority D) A random assortment of objects

Answer: B. A system requires organized interaction and a boundary (Markov Blanket), not just proximity or rules.

1. What is the function of a **Markov Blanket** in a social system like your friend group? A) It prevents any information from entering or leaving B) It filters interactions between the group's internal dynamics and the outside world C) It destroys the group over time D) It makes every member identical

Answer: B. The Markov Blanket is not a wall but a filter that mediates information flow across the boundary.

1. Your school persists as a system even though every student eventually graduates. This illustrates that systems are defined by: A) Their physical building B) Their oldest members C) Their persistent patterns of interaction D) Government funding

Answer: C. Systems maintain identity through ongoing relational patterns, not fixed parts.

1. When a new student joins a tight-knit class, both sides experience elevated **prediction error** because: A) The new student is breaking rules B) Neither side's generative model includes the other yet C) The teacher assigned too much homework D) The school system has collapsed

Answer: B. Mutual model-updating must occur before the system reaches a new equilibrium.

1. The concept of **nested systems** means that: A) Systems can only exist one at a time B) You belong to multiple overlapping systems at different scales simultaneously C) Smaller systems always control larger ones D) Only biological systems can be nested

Answer: B. You are a system within a family, within a school, within a community -- each with its own Markov Blanket.

1. A friend group that stops communicating will eventually dissolve. Active Inference explains this as: A) A failure to minimize free energy and maintain the system's boundary B) A sign that the group was never a real system C) Normal behavior for all systems D) Evidence that Markov Blankets are permanent

Answer: A. Maintaining a system requires ongoing effort against disorder; without it, the boundary dissolves.

1. When you start dating someone new, the Markov Blanket of your social system: A) Disappears entirely B) Becomes permanently fixed C) Expands and restructures to include new information flows D) Shrinks to include only two people

Answer: C. Boundaries are dynamic -- new relationships change how information flows across the system boundary.

1. Why do social transitions like changing schools feel disorienting? A) The cafeteria food is different B) Your system's Markov Blanket is being restructured, requiring extensive model-updating C) Teachers at new schools are stricter D) You lose all your memories

Answer: B. Restructuring your system boundary means your generative model must be rebuilt for a new environment.

1. Groups tend to develop shared norms and routines because: A) Leaders force everyone to conform B) Shared expectations reduce prediction error (surprise) for all members C) It is legally required D) People have no individual preferences

Answer: B. The free energy principle predicts that systems develop shared models to minimize collective surprise.

1. Which is the best example of a system maintaining its Markov Blanket? A) A pile of leaves blowing in the wind B) A sports team holding regular practices and communicating game strategy C) A random group of strangers at a bus stop D) A single rock sitting on a hill

Answer: B. The team actively maintains its boundary through organized interaction, communication, and shared routines.

Part B: Short Answer

1. Pick a system you belong to (sports team, family, friend group, online community). Describe what serves as its Markov Blanket -- what specific channels mediate information between inside and outside?
2. Explain why a group chat that goes silent for months is an example of a system failing to minimize free energy. What would it take to revive it?
3. Describe a time when someone violated an unspoken group norm. What prediction error did the group experience, and how did it respond -- by updating the model or enforcing the norm?
4. How do overlapping systems create conflicts in your daily life? For example, how might the expectations of your family system clash with those of your friend group system?
5. If you moved to a new city tomorrow, which of your current systems would survive and which would dissolve? What does this tell you about which boundaries are strongest?